# Theme and chassis layout

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Make a dimensioned layout that supports the circuit and drivetrain.

**SUCCESS CRITERIA - I can...**

1. Label component locations and clearances.
2. Use dimensions to check that parts fit.
3. Use measurements or observations to explain a structural choice.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

A **40**-turn coil has flux per turn change by **0.006 Wb** in **0.4 s**. **Find average induced-emf magnitude**.

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the prediction

**Reference:** available length = chassis length - components - clearances.
Use measured dimensions for a top view. Keep axles parallel and controls accessible. Get approval before tools/cuts.

**Worked reasoning:** A **20 cm** chassis reserves **6 cm** for the battery, **5 cm** for the motor and **2 cm** total for end clearances. The remaining **7 cm** must cover other components and gaps.

**Your turn:** 1. **Draw a dimensioned top view** showing these regions.
2. **Label any assumed dimensions rather** than pretending they were measured.




**Marzano check - circle:** ① ② ③

## 03 Read and explain the diagram or equation

**Draw a dimensioned top view** of your chassis. **Label motor**, battery, switch, axle lines and wheel clearance. **Mark the intended forward direction**.




**Marzano check - circle:** ① ② ③

## 04 Apply the physics

A sample chassis is **180 mm** long. Reserve **15 mm** at each end and a **95 mm** component region. **Mark the regions**, calculate remaining length and convert the total chassis length to meters.




**Marzano check - circle:** ① ② ③

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Continue down each column.

## 05 Explain your reasoning

Your theme adds a heavy decoration far to one side. **Predict a possible effect** and propose a way to **test it** without claiming the effect is certain.

**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

A **8 Ω** resistor is in series with two parallel **12 Ω** resistors across **28 V**. **Find parallel equivalent**, total resistance and supply current.
**Draw the stated circuit** and **label quantities relevant** to the question.

Draw the complete circuit and label shared nodes.

**Marzano check - circle:** 1 2 3

## 07 Exit check

1. **Explain why a top-view sketch** alone may miss a wheel or gear clearance problem.
2. **Add one useful side-view dimension**.




**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the prediction

A **180 mm** chassis needs two **15 mm** clearances, a **60 mm** battery and a **50 mm** motor region. **Find remaining length**. Then **name a side-view clearance** that this calculation misses.




**Marzano check - circle:** 1 2 3